

shared inputs

same-prompt  
preference pair

frozen baseline  
policy

**canonical RLHF**

**DPO**

fit scalar reward

sample  
evolving policy

PPO policy update

reward minus  
reference drift

compare policy change  
relative to reference

binary winner/loser  
loss

direct policy update

Derived later from the KL-regularized optimum  
+ Bradley–Terry; not a reduction of PPO.